Early Access Paper

AirStore is a data service that records a large number of user actions across the system and ensures they are not changed over time, or at least cannot be changed easily.

AirStore will be of significant value in the upcoming future age of robotics and artificial intelligence.

Nodes are units of computing power with data storage. They are usually run by an organisation or individual that has some real world representation as AirStore is very much focussed on recording real world actions such as financial transactions or actions by robots.

Each Node maintains a Sheet. The Sheet is a set of Entries that describe Actions undertaken on the Network. A Node will usually be run by a company or business group, like a bank or a robotics company and each Node is associated with metadata that describes its real work affiliation and the other Nodes that it trusts.

Nodes work together in a Network. Nodes can specify which Nodes they trust. If two Nodes trust each other they form a Network together. A Network is a set of Nodes in which all the Nodes trust each other. Nodes that trust each other, can request each others' public keys, this means they can authenticate each other.

All the Networks collectively are called the Ecosystem. It is important to keep in mind that not all the Nodes in the Ecosystem have to trust each other. This is what makes AirStore significantly different to other block chain like systems. It also means that many more participants can use the same technology for many different use cases without having to have any connection with each other, however since they are using the same technology, it makes future trusting relationships and the expansion of networks technically simple. This provides a framework for growing cooperation among organisations internationally without any immediate requirements to change their means of operations.

AirStore Nodes that are in a Network share the same History. Each Node's History is just the list of all the previous Entries (transactions, actions etc.) for that Network.

The process of identifying and agreeing upon the History is called Sorting.

Similar to other blockchain style systems, AirStore makes use of hashing to identify whether units of data are the same or not. The specific details of the hashing algorithms and how they are applied to ensure data remains consistent, are the subject of continuing development.

AirStore is currently in development and is intended to be a usable system, sometime in 2027.